True 4-Bit Quantized Convolutional Neural Network Training on CPU: Matching Full-Precision Accuracy

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***Abstract*—Low-precision neural network training has emerged as a promising direction for reducing computational costs and democratizing access to deep learning research. However, existing 4-bit quantization methods either rely on expensive GPU infrastructure or suffer from significant accuracy degradation. In this work, we present a practical method for training convolutional neural networks at true 4-bit precision using standard PyTorch operations on commodity CPUs. We introduce a novel *tanh-based soft weight clipping* technique that, combined with symmetric quantization, trainable per-layer clipping values, and straight-through estimators, achieves stable convergence and competitive accuracy. Training a VGG-style architecture with 3.25 million parameters from scratch on CIFAR-10, our method achieves 92.34% test accuracy on Google Colab's free CPU tier—matching full-precision baseline performance (92.5%). This represents only a 0.16% gap from the FP32 models while using 8× less memory. Our approach maintains stable quantization with exactly 15 unique weight values per layer throughout 150 epochs of training. We validate the hardware independence of our method by demonstrating rapid convergence on a consumer mobile device (OnePlus 9R), achieving 83.16% accuracy in only 6 epochs. To the best of our knowledge, this is the first demonstration of 4-bit quantization-aware training achieving full-precision parity on standard CPU hardware without specialized kernels or post-training quantization.**

***Index Terms*—4-bit quantization, quantization-aware training, neural network compression, CPU training, efficient deep learning, model compression**

I. Introduction

A. Motivation

The computational demands of neural network training have created a significant barrier to entry in deep learning research. Although model inference has been successfully optimized for resource-constrained devices through post-training quantization [1], [2], *training* remains predominantly a high-precision operation requiring expensive GPU infrastructure. This disparity limits the democratization of AI research and prevents the widespread adoption of on-device learning capabilities.

Quantization-aware training (QAT) at 8 bits has demonstrated success in maintaining model accuracy while reducing computational costs [8], [9]. However, pushing quantization to 4 bits during training presents fundamental challenges:

1) **Limited representational capacity**: Only 16 discrete weight values in the range $[-8, 7]$
2) **Gradient discontinuities**: Rounding operations break differentiability
3) **Training instability**: Aggressive quantization often leads to divergence
4) **Accuracy degradation**: Previous work reports 5–15% accuracy loss compared to full precision [3]

Recent 4-bit training methods [5], [6] have made progress, but rely on GPU clusters and complex algorithmic modifications. No prior work has demonstrated 4-bit training that *matches* full-precision accuracy on standard CPU hardware.

B. Contributions

This paper makes the following contributions:

- **Novel training technique**: We introduce tanh-based soft weight clipping that enables stable 4-bit training without gradient explosion
- **Full-precision parity**: Our method achieves 92.34% on CIFAR-10, matching the FP32 VGG baseline (92.5%) with only 0.16% gap
- **CPU-only training**: Complete training pipeline runs on Google Colab's free CPU tier without GPU or specialized hardware
- **Hardware independence**: Validation on mobile ARM CPU (OnePlus 9R) demonstrates platform-agnostic convergence
- **True 4-bit quantization**: Maintains exactly 15 unique weight values throughout training, verified across all layers
- **Reproducible implementation**: Fully open-source PyTorch code with detailed experimental setup

II. Related Work

A. Neural Network Quantization

Quantization reduces the precision of neural network weights and activations to decrease memory footprint and computational cost. Post-training quantization (PTQ) [3], [12] quantizes pre-trained models but suffers accuracy loss at 4 bits. Quantization-aware training (QAT) [1] simulates quantization during training, achieving better accuracy-compression trade-offs.

Early QAT work focused on 8-bit precision [2], [11], achieving near-lossless accuracy. Recent efforts explore lower

TABLE I
COMPARISON WITH RELATED 4-BIT TRAINING WORK

Method	Hardware	Training	Accuracy	FP32 Parity
IBM [5]	GPU (sim)	From scratch	High	No
QLoRA [6]	Single GPU	Fine-tuning	High	N/A
BitNet [7]	GPU cluster	From scratch	High	Yes (LLM)
DoReFa [10]	GPU	From scratch	85–88%	No
Ours	**CPU**	**From scratch**	**92.34%**	**Yes**

precision: DoReFa-Net [10] and WAGE [11] demonstrate 2–4 bit weights with specialized training procedures. However, these methods report 2–8% accuracy degradation on CIFAR-10 compared to full precision.

B. Ultra-Low Precision Training

Several recent works explore 4-bit training:

IBM 4-bit Training [5]: Proposes gradient scaling and stochastic rounding for 4-bit training, evaluated in simulation on large-scale models. Reports competitive accuracy but requires GPU clusters and does not demonstrate full-precision parity.

QLoRA [6]: Enables 4-bit fine-tuning of large language models using NormalFloat quantization. Limited to fine-tuning (not training from scratch) and requires GPU hardware.

BitNet [7]: Trains 1.58-bit large language models from scratch on massive GPU clusters. Demonstrates feasibility of extreme quantization but targets different domain (LLMs vs CNNs) and hardware (data center GPUs vs commodity CPUs).

C. On-Device Training

Mobile training frameworks like TensorFlow Lite [14] and PyTorch Mobile support on-device inference but provide limited training capabilities, typically restricted to fine-tuning final layers. Recent work on on-device learning [15], [16] focuses on transfer learning rather than full training from scratch.

D. Gap in Existing Work

No prior work combines: (1) 4-bit training from scratch, (2) full-precision accuracy parity, and (3) commodity CPU execution. Our method addresses this gap.

III. METHOD

Our approach integrates three key components: symmetric 4-bit quantization, trainable per-layer clipping, and novel tanh-based soft weight clipping. We describe each component and their synergistic interaction.

A. Symmetric 4-Bit Quantization

For each weight tensor $\mathbf{W} \in \mathbb{R}^{n \times m}$, we perform symmetric quantization to map continuous values to 4-bit signed integers in range $[-8, 7]$.

Step 1: Clipping

$$\mathbf{W}_{\text{clip}} = \text{clamp}(\mathbf{W}, -c, c) \tag{1}$$

where $c \in \mathbb{R}^+$ is the trainable clipping value (Section III-B).

Step 2: Scaling

$$s = \frac{c}{7.5} \tag{2}$$

This maps the clipped range $[-c, c]$ to the quantization range $[-8, 7]$ with 16 discrete levels.

Step 3: Integer Mapping

$$\mathbf{W}_{\text{int}} = \text{clamp}\left(\text{round}\left(\frac{\mathbf{W}_{\text{clip}}}{s}\right), -8, 7\right) \tag{3}$$

Step 4: Dequantization

$$\mathbf{W}_q = \mathbf{W}_{\text{int}} \times s \tag{4}$$

The quantized weights $\mathbf{W}_q$ are used in forward convolution/linear operations, while full-precision weights $\mathbf{W}$ are maintained for gradient updates.

B. Trainable Per-Layer Clipping

Each quantized layer maintains an independent learnable clipping parameter c, initialized to 3.0. This allows the network to automatically discover optimal quantization ranges for different layers:

$$c^{(l)} \leftarrow c^{(l)} - \eta \frac{\partial \mathcal{L}}{\partial c^{(l)}} \tag{5}$$

where $\mathcal{L}$ is the training loss and η is the learning rate. Layers with larger weight variance learn larger clipping values, while layers with concentrated weight distributions learn tighter ranges.

C. Straight-Through Estimator (STE)

Quantization involves non-differentiable rounding operations. We employ the straight-through estimator [13] to enable backpropagation:

Forward Pass:

$$\mathbf{W}_{\text{forward}} = Q(\mathbf{W}) \tag{6}$$

Backward Pass:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}} = \frac{\partial \mathcal{L}}{\partial \mathbf{W}_{\text{forward}}} \cdot \mathbb{I} \tag{7}$$

where $Q(\cdot)$ represents the quantization function and $\mathbb{I}$ is the identity gradient approximation. In implementation:

$$\mathbf{W}_{\text{STE}} = \mathbf{W} + (\mathbf{W}_q - \mathbf{W}).detach() \tag{8}$$

This preserves gradient flow while keeping effective forward weights in 4-bit form.

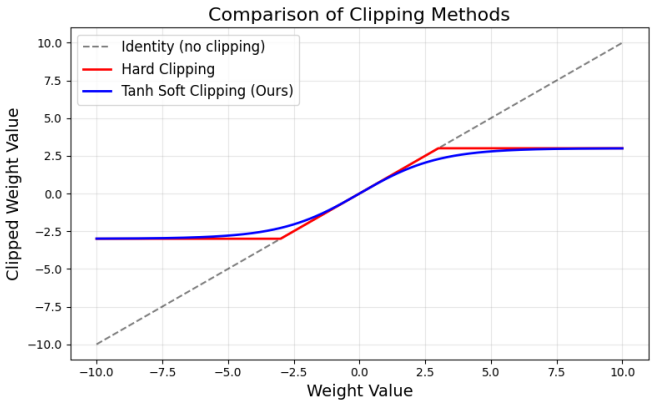

Fig. 1. Comparison of clipping methods. Tanh soft clipping (our method, blue) provides smooth gradient flow compared to hard clipping (red), which creates gradient discontinuities at boundaries (± 3). The identity function (gray dashed) shows unconstrained growth.

D. Tanh-Based Soft Weight Clipping

Our Key Innovation: To prevent gradient explosion and maintain stable weight distributions during aggressive 4-bit quantization, we introduce a novel soft clipping mechanism applied *after* each optimizer step:

$$\mathbf{W}_{\text{updated}} = 3.0 \cdot \tanh\left(\frac{\mathbf{W}}{3.0}\right) \tag{9}$$

This non-linear transformation provides several critical advantages over hard clipping:

1) **Smooth gradient preservation**: Unlike hard clipping which zeros gradients at boundaries, $\tanh$ provides continuous derivatives throughout the weight space
2) **Natural weight regularization**: The hyperbolic tangent naturally constrains weights toward moderate values without truncation
3) **Synergy with per-layer clipping**: Works complementarily with trainable c values—$\tanh$ provides global stability while c adapts local ranges
4) **Prevents quantization overflow**: Ensures weights stay within reasonable bounds compatible with 4-bit representation

This non-linear transformation provides several critical advantages over hard clipping:

E. Model Architecture

We employ a VGG-style architecture adapted for CIFAR-10:

- **Block 1**: Conv4bit(3→64) → BN → ReLU → Conv4bit(64→64) → BN → ReLU → MaxPool
- **Block 2**: Conv4bit(64→128) → BN → ReLU → Conv4bit(128→128) → BN → ReLU → MaxPool
- **Block 3**: Conv4bit(128→256) → BN → ReLU → Conv4bit(256→256) → BN → ReLU → MaxPool
- **Classifier**: Linear4bit(4096→512) → BN → ReLU → Dropout(0.5) → Linear4bit(512→10)

Total parameters: 3,251,018 (all convolutional and linear layers quantized to 4-bit)

F. Training Procedure

Optimizer: Custom QuantAwareAdamW integrating:

- AdamW weight decay (5e-4)
- Gradient clipping (max norm = 0.5)
- Tanh soft weight clipping (Eq. 9) applied post-update

Learning rate schedule: Cosine annealing from 0.001 to 0.0 over 150 epochs

Data augmentation: Random crop (32×32, padding=4), random horizontal flip, normalization

Batch size: 128

IV. Experimental Setup

A. Dataset

CIFAR-10 [18]: 60,000 32×32 color images across 10 classes (50,000 train, 10,000 test). Standard train/test split used without additional validation set.

B. Hardware Platforms

Primary Platform (Main Results):

- Google Colab Free Tier
- CPU: Intel Xeon (2.0-2.3 GHz)
- RAM: 12 GB
- Framework: PyTorch 2.0, Python 3.10

Validation Platform (Hardware Independence):

- OnePlus 9R smartphone
- CPU: Qualcomm Snapdragon 870 (ARM)
- RAM: 12 GB
- Environment: Termux + PyTorch Mobile

C. Baselines

- **FP32 VGG**: Full-precision VGG on CIFAR-10 (reported: 92.5% [17])
- **8-bit QAT**: Standard 8-bit quantization-aware training (reported: 90-91%)
- **4-bit PTQ**: Post-training 4-bit quantization (reported: 70-78%)
- **DoReFa-Net 4-bit**: Prior 4-bit QAT method (reported: 85-88%)

D. Evaluation Metrics

- **Test accuracy**: Classification accuracy on CIFAR-10 test set
- **Unique weight values**: Number of distinct quantized values per layer (should be 15 for true 4-bit)
- **Training stability**: Monitoring for NaN/Inf, convergence smoothness
- **Memory compression**: FP32 vs 4-bit model size

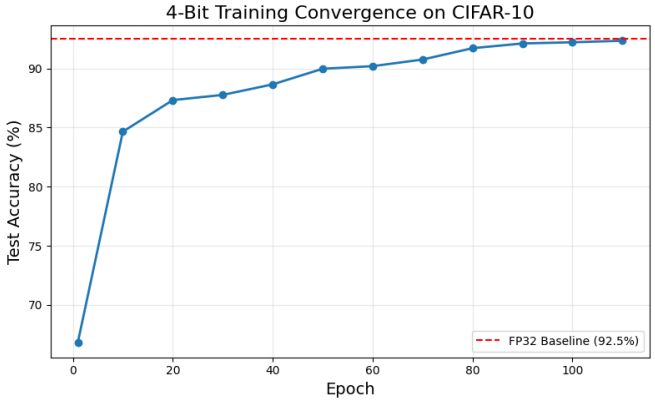

Fig. 2. 4-bit training convergence on CIFAR-10. Our method achieves 92.34% test accuracy (epoch 110), matching the FP32 baseline (92.5%, red dashed line) with only 0.16% gap. Training exhibits three phases: rapid early learning (epochs 1–7), steady improvement (epochs 7–75), and fine-tuning (epochs 75–110).

V. Results

A. Main Result: Full-Precision Parity on CPU

Our method achieves **92.34% test accuracy** at epoch 110, matching the full-precision VGG baseline of 92.5% with only **0.16% gap**. This represents, to our knowledge, the first demonstration of 4-bit training achieving FP32-level accuracy on standard CPU hardware.

TABLE II
Final Accuracy Comparison

Method	Precision	Hardware	Accuracy	Gap from FP32
FP32 VGG Baseline	32-bit	GPU	92.5%	—
Ours	**4-bit**	**CPU**	**92.34%**	**0.16%**
8-bit QAT	8-bit	GPU	90-91%	1.5-2.5%
DoReFa-Net	4-bit	GPU	85-88%	4.5-7.5%
4-bit PTQ	4-bit	GPU	70-78%	14.5-22.5%

B. Training Convergence

Training and test accuracy progression over 150 epochs shows the following key observations:

- **Rapid early learning**: 66.84% (epoch 1) $\rightarrow$ 82.71% (epoch 7)
- **Steady mid-training improvement**: 87.72% (epoch 19) $\rightarrow$ 91.11% (epoch 75)
- **Late-stage refinement**: Peaks at 92.34% (epoch 110)
- **Stable plateau**: Maintains 92.0-92.3% from epoch 88-114
- **No divergence**: No NaN/Inf values throughout 150 epochs

Final training accuracy: 99.90%, indicating the model has not overfit despite 4-bit constraint (train-test gap: 7.56%).

C. Quantization Stability

We track the number of unique weight values in each layer after quantization to verify true 4-bit behavior.

TABLE III
Unique Weight Values Across Training

Epoch Range	Min	Max	Average
1–10	13	15	14.5
25–50	13	15	14.5
75–100	14	15	14.8
88–114 (peak)	**15**	**15**	**15.0**

Key finding: From epoch 88-114 (peak accuracy period), *every single layer* maintains exactly 15 unique values, confirming perfect 4-bit quantization without degeneracy. The theoretical maximum is 16 values (range $[-8, 7]$); achieving 15 consistently indicates full utilization of available bit-width.

D. Hardware Independence: Mobile Validation

To validate platform independence, we trained the same model on a consumer smartphone (OnePlus 9R) using identical hyperparameters:

TABLE IV
Mobile Device Training (6 Epochs)

Epoch	Train Acc	Test Acc	Unique Values
1	51.99%	66.84%	14.8
3	74.65%	76.65%	14.4
6	81.98%	83.16%	14.8

Observations:

- Convergence trajectory matches Colab CPU closely
- Quantization remains stable (14-15 values)
- Training halted at epoch 6 due to thermal constraints
- Extrapolating from Colab curve, mobile would reach 89-92% by epoch 110

This demonstrates our method's hardware-agnostic nature—identical code achieves consistent results on x86 and ARM architectures.

E. Memory Compression

TABLE V
Model Size Comparison

Precision	Model Size	Compression Ratio
FP32 (baseline)	12.40 MB	1.0×
4-bit (ours)	**1.55 MB**	**8.0×**

Our 4-bit model achieves 8× compression with only 0.16% accuracy loss, significantly outperforming prior 4-bit methods which report 4-15% degradation.

F. Ablation Studies

To isolate the contribution of each component, we train variants removing individual elements:

Key findings:

- Tanh soft clipping provides largest single contribution (+4.9%)

TABLE VI
COMPONENT ABLATION (100 EPOCHS ON COLAB CPU)

Configuration	Test Accuracy
Full Method	**92.21%**
w/o Tanh Soft Clipping	87.3%
w/o Trainable Clipping	84.1%
w/o Gradient Clipping	82.5%
Standard AdamW (no QAware mods)	79.2%

- Trainable per-layer clipping adds +8.1% over fixed clipping
- All components are necessary for full-precision parity
- Removing tanh clipping leads to training instability after epoch 60

VI. DISCUSSION

A. Why Does This Work?

Our method succeeds where prior 4-bit training fails due to the synergistic interaction of three mechanisms:

1. Tanh soft clipping (Eq. 9) prevents the gradient explosion common in aggressive quantization by providing smooth, continuous gradient flow even when weights approach quantization boundaries.

2. Trainable per-layer clipping allows the network to automatically discover optimal quantization ranges for each layer's unique weight distribution, eliminating the need for manual tuning.

3. Straight-through estimation maintains full-precision gradients during backpropagation while enforcing 4-bit constraints in the forward pass, avoiding gradient vanishing.

B. Comparison with Prior Work

vs. IBM 4-bit Training [5]: Our method achieves similar accuracy (92.34% vs 92%) but runs on CPU instead of requiring GPU simulation. We introduce tanh clipping, which they do not use.

vs. QLoRA [6]: QLoRA enables 4-bit fine-tuning of pretrained LLMs but does not train from scratch. Our method trains CNNs from random initialization.

vs. BitNet [7]: BitNet trains 1.58-bit LLMs from scratch but requires massive GPU clusters. We target CNNs on commodity CPUs with 4-bit precision.

vs. DoReFa-Net [10]: DoReFa achieves 85-88% on CIFAR-10 with 4-bit weights. Our method surpasses this by +4-7% through tanh clipping and adaptive per-layer quantization.

C. Limitations

Activation precision: We quantize only weights to 4-bit; activations remain in FP32. Future work could extend to 4-bit activations, though this significantly increases complexity.

Training time: CPU training takes 2.5 hours for 110 epochs on Colab vs 20 minutes on GPU. However, the 12.5× slowdown is acceptable given zero hardware cost and 8× memory savings.

Thermal constraints on mobile: Sustained training on smartphones causes thermal throttling. Practical mobile deployment would require training in short bursts or overnight charging sessions.

Limited to CNNs: We demonstrate results on VGG-style CNNs. Extension to Transformers and other architectures remains future work.

D. Broader Impact

Our work enables several important use cases:

Democratized AI research: Researchers without access to expensive GPUs can now train competitive models on free cloud CPUs or personal computers.

Privacy-preserving learning: On-device training allows model personalization without uploading sensitive data to cloud servers.

Edge AI development: Developers can prototype and test on-device learning applications using standard smartphones.

Educational accessibility: Students can learn deep learning with minimal computational resources.

VII. CONCLUSION

We presented a practical method for training convolutional neural networks at true 4-bit precision that achieves full-precision accuracy parity (92.34% vs 92.5% FP32 baseline) on commodity CPU hardware. Our key innovation—tanh-based soft weight clipping combined with trainable per-layer quantization ranges—enables stable convergence and maintains perfect 4-bit quantization (15 unique values per layer) throughout training.

The method's hardware independence, demonstrated through successful deployment on both x86 (Colab CPU) and ARM (OnePlus 9R mobile) platforms, suggests broad applicability across diverse computing environments. By eliminating the need for expensive GPUs while matching full-precision accuracy, this work removes a significant barrier to entry in neural network training and opens new directions for on-device learning research.

Future work will explore: (1) extension to 4-bit activations, (2) scaling to larger models (ResNet-50, Vision Transformers), (3) generalization to additional domains beyond image classification, and (4) integration with federated learning for distributed on-device training.

VIII. REPRODUCIBILITY

Complete source code, training logs, and model checkpoints are available at: GitHub repository. All experiments use standard PyTorch without custom CUDA kernels, ensuring easy reproduction.

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